

DxPS

Real-time Provisioning System

Test Cases, Test Results, Security Verification and Payload Evidence

Go + Apache Kafka 4 (KRaft) + PostgreSQL 18, exercised against mTLS HTTP/2 network element simulators

177/177	91.5%	24/24	0
automated tests passed	statement coverage	live security probes	known CVEs (govulncheck)

Item	Value
Document	DXPS-TEST-CASES-RESULTS
Version	1.0
Test run	06 October 2026 21:12 Pacific Daylight Time (suite wall time 77.2 s)
Design baseline	DxPS_TO-BE_HLD_LLD.docx v1.1
System under test	dxps Go module: 6 services (gateway, orchestrator, adapter, eventhub, netsim, dashboard), dxpsctl, 24 internal packages
Environment	Windows 11 (10.0.26200), Go 1.26.6, Apache Kafka 4.3.1 (KRaft, single node), PostgreSQL 18.6 (TLS 1.3, SCRAM-SHA-256, RLS), python-docx report generator
Source of every number	Generated from go test -json, the merged coverage profile, the live probe run, govulncheck, gosec and captured simulator exchanges (no hand-typed results)

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1. Executive summary

All 177 automated tests pass (173 documented test cases plus their sub-tests). The suite covers 91.5% of the 4,196 statements in the DxPS internal packages, measured across package boundaries so integration tests count toward the code they exercise. All 24 live security probes pass against the running stack, govulncheck reports no reachable vulnerabilities, and gosec reports no high or medium findings.

Testing found and fixed eight defects, six of them security-relevant (section 7). The most important: the simulator exchange log exposed OAuth access tokens on the dashboard; the dashboard accepted any Host header (DNS rebinding); the event hub followed redirects as retries; and a missing dependency result was rendered as JSON null into an NE payload instead of failing the task.

142	23	7	54	173/173
unit	integration	end-to-end	security-focused	documented TCs passed

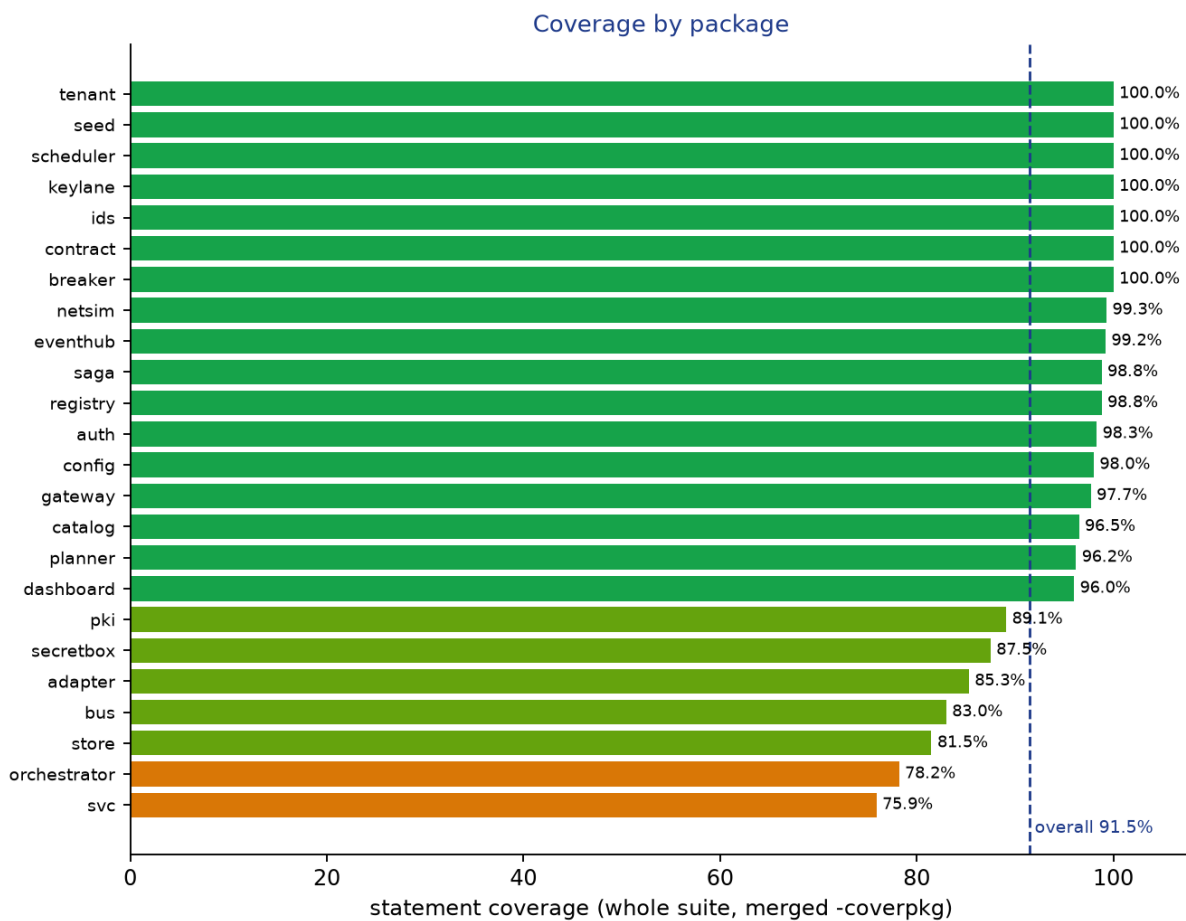


Figure 1 - Statement coverage per package (whole suite)

2. Scope and test approach

Level	What it proves	How it runs
Unit	Pure logic: JWT, tenant ids, catalog/CEL, planner DAGs, DRR scheduler, key lanes, saga, breaker, PKI	go test, no external dependency
Component	Each service handler against network simulators over HTTP/2 mTLS (adapters, event hub, gateway)	httptest servers, netsim in-process

Integration	Real PostgreSQL 18 (RLS, roles, outbox, migrations) and Kafka 4 (tiered consumers, exactly-once, retry ladder)	local runtime; isolated zz-test-* tenants and uniquely named topics
End-to-end	TMF641 order -> gateway -> PostgreSQL -> orchestrator -> planner -> adapter -> simulator -> result -> TMF688 event, including compensation and retries	orchestrator harness with real gateway and store
Security	24 live attack probes, RLS and least-privilege checks, injection, SSRF, CSRF, DNS rebinding, mTLS, token forgery, secret redaction	tests + mosaic probe run + govulncheck + gosec
Load	2000 mixed orders at 100/s with 10% chaos faults through the running stack	mosaic load generator

Network simulators

Every southbound call goes to a simulator, never to a stub inside the adapter. Simulators speak HTTP/2 over mutual TLS, validate payloads against the interface rules, keep per-tenant state and record each exchange (secrets redacted). Faults are injected deterministically through the subscriber identifier so a test or load run can target an exact behaviour:

Trigger	Simulator behaviour	DxPS behaviour under test
MSISDN/EID ends 997	two HTTP 503, then success	retry ladder (dxps.retry.* topics) recovers; order completes
ends 998	permanent 400 (SM-DP+: ES2+ Failed inside HTTP 200)	task fails, ATOMIC saga compensates in reverse
ends 999	HTTP 503 every time	bounded retries, DXPS-4001 after MaxAttempts, rollback
USP device OFFLINE	USP error 7002	permanent failure mapped to a DXPS code
QoD bandwidth > 10000	HTTP 409	business rejection surfaced to the order
admin API	latency and random error rate per simulator	chaos/latency experiments from the mosaic

3. Test environment

Component	Version / setting
Go toolchain	go1.26.6 windows/amd64 (go.mod: go 1.26.5, toolchain go1.26.6)
Apache Kafka	4.3.1, KRaft combined broker+controller, 127.0.0.1:9092, 41 DxPS topics, KIP-848 consumer groups
PostgreSQL	18.6, port 5433, TLS 1.3 only, SCRAM-SHA-256; roles dxps_owner (DDL), dxps_app (RLS enforced), dxps_ops (BYPASSRLS, registry/relay only)
Services	gateway https://127.0.0.1:8443 (TLS), orchestrator, adapter, eventhub, netsim (NEs on 9101-9108 mTLS, webhook sink 9109), mosaic dashboard http://127.0.0.1:8088 (loopback only)
Tenants	host-mno (host MNO), mvno-alpha (full MVNO, own OCS), mvno-beta (light MVNO), ent-acme (enterprise), mvno-gamma (suspended); tests use zz-test-* tenants for isolation
Test isolation	integration tests create uniquely named topics/groups and never delete topics; test rows are marked published so the live relay ignores them

4. Results by package

Package	Component	Passed	Coverage	Result
adapter	Southbound adapters (SBA, IMS, NETCONF/USP, eSIM, BSS, NPDB, CAMARA)	17/17	85.3%	PASS
auth	Gateway security: JWT authentication and scope authorisation	9/9	98.3%	PASS
breaker	Per-NE circuit breaker	2/2	100.0%	PASS
bus	Kafka bus: tiered consumers, EOS, retry ladder	7/7	83.0%	PASS
catalog	Command catalog: specs, CEL rules, payload templates	8/8	96.5%	PASS
config	Configuration and secret loading	7/7	98.0%	PASS
contract	Kafka contracts, topics and keys	4/4	100.0%	PASS
dashboard	Mosaic dashboard and live security probes	13/13	96.0%	PASS
eventhub	TMF688 event hub (webhooks, HMAC, retries, SSRF guard)	8/8	99.2%	PASS
gateway	TMF641 northbound gateway	16/16	97.7%	PASS
ids	Identifiers: UUIDv7 and deterministic derived ids	3/3	100.0%	PASS
keylane	Entity key lanes (per-subscriber ordering)	3/3	100.0%	PASS
netsim	Network element simulators (9 NEs + webhook sink)	22/22	99.3%	PASS
orchestrator	Orchestrator: sagas, dependencies, sequencing (E2E)	13/13	78.2%	PASS

pki	PKI and mutual TLS	4/4	89.1%	PASS
planner	Planner: business command to NE task DAG	7/7	96.2%	PASS
registry	NE registry, shared-NE access and quotas	6/6	98.8%	PASS
saga	Saga compensation and retry back-off	5/5	98.8%	PASS
scheduler	Deficit round-robin fair scheduler (tenant x priority)	5/5	100.0%	PASS
secretbox	Secret encryption at rest (AES-256-GCM)	3/3	87.5%	PASS
seed	Reference data (tenants, NEs)	0/0	100.0%	n/a
store	PostgreSQL store: RLS, outbox, migrations, seed	8/8	81.5%	PASS
svc	Service bootstrap, logging, HTTP server hardening	3/3	75.9%	PASS
tenant	Tenant registry, identifiers and SLA weights	4/4	100.0%	PASS

Counts include sub-tests (t.Run). Coverage is measured with `-coverpkg=./internal/...` and merged across test binaries: a statement counts as covered if any test in the suite executes it. The seed package holds reference data only and has no tests of its own.

Where coverage is below 90% and why

- orchestrator (78%)**: the remaining statements are database-error branches inside multi-step transactions (for example, a failure between planning and dispatch) that need fault injection inside PostgreSQL to reach.
- store (82%), bus (83%)**: error returns from PostgreSQL/Kafka calls that do not fail on a healthy local cluster; the success paths, RLS and failure propagation are covered.
- adapter (85%)**: the Kafka consumer wiring of the adapter service; driver logic for all seven domains is covered against the simulators.
- svc (76%), pki (89%), secretbox (88%)**: os.Exit paths and errors from crypto/rand, which cannot fail in practice.

5. Test case catalogue

Each test case is an automated Go test. The id and description come from the comment above the test function, so this catalogue cannot drift from the code. The result and duration come from this run.

5.1 adapter - Southbound adapters (SBA, IMS, NETCONF/USP, eSIM, BSS, NPDB, CAMARA)

Design reference: LLD 6.8 Adapters. Source: `internal/adapter/adapter_test.go`. Coverage 85.3%.

ID	Level	Objective and expected result	Go test	Result
TC-ADP-001	Unit / Sec	SBA driver fetches an NRF OAuth2 token, sets 3GPP SBI headers and provisions UDR auth data over HTTP/2 mTLS.	TestSBAProvision	PASS 0.03s
TC-ADP-002	Unit	UDR validation errors are FAILED with the 3GPP cause mapped into the DxPS error code.	TestSBAValidationFailure	PASS 0.01s
TC-ADP-003	Unit / Sec	IMS ordering precondition and tenant isolation on the shared IMS-PG.	TestIMS	PASS 0.02s
TC-ADP-004	Unit	RESTCONF YANG-Patch to the PE router: success, capacity rejection (409) and site delete.	TestNetconf	PASS 0.01s
TC-ADP-005	Unit	yang-patch-status without "ok" in a 2xx response is a FAILED outcome.	TestYangPatchStatus	PASS 0.00s
TC-ADP-006	Unit	access domain: ONT activation via TR-385 YANG-Patch and CPE config via USP; offline agent fails (7002).	TestAccess	PASS 0.02s
TC-ADP-007	Unit	eSIM ES2+ success and business failures reported inside HTTP 200 are interpreted as FAILED.	TestESIM	PASS 0.02s
TC-ADP-008	Unit	BSS account (TMF666, X-Request-ID = idempotency key), product, NPDB port and CAMARA QoD session.	TestBSSAndExposure	PASS 0.02s
TC-ADP-009	Unit	NE 503 with Retry-After is RETRYABLE and carries the delay; 997 recovers on the third attempt.	TestRetryable	PASS 0.02s
TC-ADP-010	Unit	SPI guard rails: unsupported op, unknown NE, not entitled, bad/missing path parameters (path traversal).	TestGuards	PASS 0.00s
TC-ADP-011	Unit	circuit breaker opens after consecutive transport failures and short-circuits with RETRYABLE (5 s).	TestBreakerAndTransport	PASS 0.01s
TC-ADP-012	Unit / Sec	client without a certificate cannot reach mTLS simulators (transport failure, RETRYABLE).	TestNoClientCert	PASS 0.01s
TC-ADP-013	Unit / Sec	NRF failures (no NRF, NRF error status, invalid token response) make SBA tasks RETRYABLE.	TestNRFFailures	PASS 0.01s
TC-ADP-014	Unit	NE without endpoints and a malformed base URI fail cleanly; quota wait honours context cancel.	TestEndpointErrors	PASS 0.00s
TC-ADP-015	Unit	problem() maps 3GPP ProblemDetails, CAMARA and TMF630 errors to DXPS codes; timeouts per tier.	TestProblemAndTimeout	PASS 0.00s
TC-ADP-016	Unit	Process validates the Kafka record (decode, tenant header/key/payload match) and emits a TaskResult.	TestProcess	PASS 0.01s

TC-ADP-017	Unit / Sec	stats endpoint binds to loopback only and serves per-NE stats.	TestServeStats	PASS 0.05s
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5.2 auth - Gateway security: JWT authentication and scope authorisation

Design reference: LLD 7.1 Security. Source: internal/auth/jwt_test.go. Coverage 98.3%.

ID	Level	Objective and expected result	Go test	Result
TC-AUTH-001	Unit / Sec	A token issued by DxPS verifies and carries tenant, subject and scopes.	TestIssueVerifyRoundTrip	PASS 0.00s
TC-AUTH-002	Unit	Signing keys shorter than 256 bits are rejected.	TestShortKeyRejected	PASS 0.00s
TC-AUTH-003	Unit / Sec	Tokens cannot be issued for an invalid tenant identifier (SQL/charset injection).	TestIssueInvalidTenant	PASS 0.00s
TC-AUTH-004	Unit / Sec	Algorithm confusion is blocked (alg=none, RS256, HS512, wrong typ).	TestAlgorithmConfusion	PASS 0.00s
TC-AUTH-005	Unit / Sec	Any modification of the claims (e.g. tenant swap) invalidates the signature.	TestTamperedClaims	PASS 0.00s
TC-AUTH-006	Unit	Expiry and not-before are enforced with a 30 s leeway.	TestTimeClaims	PASS 0.00s
TC-AUTH-007	Unit	Issuer, audience, missing exp and invalid tenant claims are rejected.	TestClaimValidation	PASS 0.00s
TC-AUTH-008	Unit / Sec	Malformed tokens (empty, oversized, wrong segment count, bad base64/JSON) are rejected.	TestMalformed	PASS 0.00s
TC-AUTH-009	Unit	Bearer extraction is case-insensitive on the scheme and rejects other schemes.	TestBearerToken	PASS 0.00s

5.3 breaker - Per-NE circuit breaker

Design reference: LLD 6.6 Resilience. Source: internal/breaker/breaker_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
TC-BRK-001	Unit	The breaker opens after N consecutive failures and rejects calls during cool-down.	TestOpens	PASS 0.00s
TC-BRK-002	Unit	After cool-down exactly one probe is admitted; its result closes or re-opens the breaker.	TestHalfOpen	PASS 0.00s

5.4 bus - Kafka bus: tiered consumers, EOS, retry ladder

Design reference: LLD 6.1 Kafka. Source: internal/bus/bus_test.go. Coverage 83.0%.

ID	Level	Objective and expected result	Go test	Result
TC-BUS-001	Unit	topic layout: 41 topics (4 BC tiers, 28 task topics, result, 3 retry, DLQ, 3 compacted state, event).	TestTopics	PASS 0.00s
TC-BUS-002	Unit	headers, records and outbox conversion; tier routing of the producer.	TestRecordsAndHeaders	PASS 0.00s
TC-BUS-003	Integration	idempotent tier producers publish P0/P1 (fast) and P2/P3 (bulk) with the producer header.	TestProducerIntegration	PASS 0.16s
TC-BUS-004	Integration	tiered KIP-848 consumer delivers every record once, keeps per-key order, reports handler errors via OnError and commits offsets (a restarted consumer in the same group does not see committed records again).	TestTieredConsumerIntegration	PASS 6.75s
TC-BUS-005	Integration	consume-process-produce in transactions: every input yields exactly one output visible to read_committed consumers; restarting the session does not reprocess committed input.	TestEOSIntegration	PASS 18.47s
TC-BUS-006	Integration	retry forwarder holds a record until not_before, strips ladder headers and republishes it to its target topic; records without a target go to the DLQ topic.	TestRetryForwarderIntegration	PASS 1.63s
TC-BUS-007	Integration	admin: topic creation is idempotent; end offsets cover all topics; group lag is reported.	TestAdminIntegration	PASS 0.09s

5.5 catalog - Command catalog: specs, CEL rules, payload templates

Design reference: LLD 6.3 Catalog. Source: internal/catalog/catalog_test.go. Coverage 96.5%.

ID	Level	Objective and expected result	Go test	Result
TC-CAT-001	Unit	Embedded catalog loads; every spec compiles (CEL), templates parse and examples validate and render to JSON.	TestTC_CAT_001_EmbeddedCatalogValid	PASS 0.03s
TC-CAT-002	Unit	Tenant override resolution - mvno-alpha gets its own spec, other tenants get GLOBAL; unknown spec/version rejected (DXPS-1001).	TestTC_CAT_002_TenantResolution	PASS 0.01s
TC-CAT-003	Unit	Parameter validation - required, unknown (additionalProperties=false), type, pattern, enum, range, length.	TestTC_CAT_003_ParamValidation	PASS 0.01s
TC-CAT-004	Unit / Sec	Template rendering escapes injected JSON (no payload injection) and rejects invalid output.	TestTC_CAT_004_TemplateEscaping	PASS 0.01s
TC-CAT-005	Unit	Malformed specs are rejected at load (unknown fields, bad CEL, bad types, unknown templates, cycles).	TestTC_CAT_005_InvalidSpecsRejected	PASS 0.01s

TC-CAT-006	Unit	CEL evaluation - when conditions, coalesce keys and entity key errors.	TestTC_CAT_006_CEL	PASS 0.00s
TC-CAT-007	Unit	A value taken from a completed dependency (eSIM download -> iccid) must be present: a missing or empty value fails the render instead of sending JSON null to the NE.	TestTC_CAT_007_RequiredDependencyValue	PASS 0.01s
TC-CAT-008	Unit	Spec versions compare numerically per segment (1.9 < 1.10, 1.0 < 1.0.1), so the newest active version is selected; parameter error and default helpers.	TestVersionLess	PASS 0.00s

5.6 config - Configuration and secret loading

Design reference: LLD 8 Deployment. Source: internal/config/config_test.go. Coverage 98.0%.

ID	Level	Objective and expected result	Go test	Result
TC-CFG-001	Unit / Sec	Secrets load from the runtime env file; explicit environment variables win; lists are trimmed.	TestLoad	PASS 0.02s
TC-CFG-002	Unit	Missing, malformed or short keys are fatal configuration errors.	TestKeyErrors (+4 sub-tests)	PASS 0.04s
TC-CFG-003	Unit	an env file line with an empty key is rejected (the key, never the value, is named in the error).	TestEnvFileBadKey	PASS 0.02s

5.7 contract - Kafka contracts, topics and keys

Design reference: LLD 6.1 Kafka topology. Source: internal/contract/contract_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
TC-CON-001	Unit	Priorities parse from TMF ("0", "3", "P2") and render as topic suffixes.	TestPriority	PASS 0.00s
TC-CON-002	Unit	Topic naming and retry-ladder selection by delay.	TestTopics	PASS 0.00s
TC-CON-003	Unit / Sec	Header tenant, key prefix and payload tenant must agree (cross-tenant message injection).	TestCheckTenant	PASS 0.00s
TC-CON-004	Unit	BusinessCommand validation and entity keys.	TestValidate	PASS 0.00s

5.8 dashboard - Mosaic dashboard and live security probes

Design reference: Operations. Source: internal/dashboard/dashboard_test.go. Coverage 96.0%.

ID	Level	Objective and expected result	Go test	Result
TC-DSH-001	Unit	strict CSP and hardening headers on static assets and API; API responses are not cached.	TestHeaders	PASS 0.20s
TC-DSH-002	Unit / Sec	DNS-rebinding guard: requests with a foreign Host header are refused (421).	TestHostGuard	PASS 0.01s
TC-DSH-003	Unit / Sec	mutating endpoints require allow-listed Origin, matching HMAC-bound CSRF cookie+header and JSON.	TestCSRF	PASS 0.03s
TC-DSH-004	Unit	demo and fault request bounds; 4 KiB body cap on mutating endpoints.	TestRequestBounds	PASS 0.01s
TC-DSH-005	Unit / Sec	fault requests are proxied to the netsim admin API with the admin token; unreachable netsim -> 502.	TestFaultProxy	PASS 0.03s
TC-DSH-006	Unit / Sec	the dashboard (which can mint tokens) only binds to loopback.	TestListenLoopback	PASS 0.04s
TC-DSH-007	Unit / Sec	gateway client trusts only the given CA and requires TLS 1.3.	TestGatewayClient	PASS 0.02s
TC-DSH-008	Unit	every generated demo order passes the real gateway validation (catalog, params, CEL, dependsOn).	TestDemoOrdersValid	PASS 2.91s
TC-DSH-009	Unit	chaos suffixes, tenant weights, counters and helpers.	TestDemoHelpers	PASS 0.01s
TC-DSH-010	Integration / Sec	all 24 live security probes pass against a real gateway (PostgreSQL RLS) and an mTLS NE.	TestSecurityProbesIntegration	PASS 3.59s
TC-DSH-011	Integration	load generator sends the requested number of orders at the requested rate and registers one webhook per tenant; a second start while running is refused (409).	TestDemoRunIntegration	PASS 0.79s
TC-DSH-012	Integration / Sec	snapshot aggregates PostgreSQL, loopback stats, netsim and test results; cached ~1 s; unavailable sources are reported as down instead of failing the snapshot.	TestSnapshotIntegration	PASS 1.70s
TC-DSH-013	Unit / Sec	the mTLS probe passes only on a real server-side refusal. A server that does not demand client certificates, an untrusted simulator, a silent listener or a bad target all fail the probe; genuine refusals (no certificate / foreign CA) pass repeatedly, whether the OS surfaces them as a TLS alert or a reset.	TestMTLSProbeVerdicts	PASS 3.47s

5.9 eventhub - TMF688 event hub (webhooks, HMAC, retries, SSRF guard)

Design reference: LLD 6.9 Events. Source: internal/eventhub/eventhub_test.go. Coverage 99.2%.

ID	Level	Objective and expected result	Go test	Result
TC-HUB-001	Unit / Sec	HMAC-SHA256 signatures verify; tampered body, wrong secret, stale/future timestamps and malformed headers fail.	TestSignVerify	PASS 0.00s
TC-HUB-002	Unit	TMF688 listener query filter on eventType and state (comma lists, unknown keys reject).	TestMatches	PASS 0.00s
TC-HUB-003	Unit / Sec	callback allow-list: https only, host:port must be allow-listed, no userinfo (SSRF guard).	TestAllowed	PASS 0.00s
TC-HUB-004	Unit	delivery signs the body, sets tenant/event headers, retries 5xx/429 with backoff, stops on 4xx and never follows redirects.	TestDeliver	PASS 0.10s
TC-HUB-005	Unit / Sec	delivery stops when the context is cancelled and reports transport errors / bad URLs.	TestDeliverCancelAndErrors	PASS 0.26s
TC-HUB-006	Unit	delivery statistics count successes/failures and keep the last 30 deliveries.	TestStats	PASS 0.00s
TC-HUB-007	Unit	Decode restores the tenant from the Kafka header (never trusted from the payload).	TestDecode	PASS 0.00s
TC-HUB-008	Integration / Sec	Handle loads the tenant's hubs under RLS, filters by query, decrypts the per-tenant secret, refuses non-allow-listed callbacks and rejects tenant mismatches.	TestHandleIntegration	PASS 0.33s

5.10 gateway - TMF641 northbound gateway

Design reference: LLD 6.2 Northbound API. Source: internal/gateway/gateway_test.go. Coverage 97.7%.

ID	Level	Objective and expected result	Go test	Result
TC-GW-001	Unit / Sec	security headers on every response; /healthz is unauthenticated.	TestSecurityHeaders	PASS 0.03s
TC-GW-002	Unit / Sec	authentication: missing, malformed, forged, expired and wrong-audience tokens are 401 with WWW-Authenticate.	TestAuthentication	PASS 0.01s
TC-GW-003	Unit / Sec	authorization: scopes per operation, X-Tenant needs dxps:tenant:any, unknown/suspended tenants are 403.	TestAuthorization	PASS 0.02s
TC-GW-004	Unit / Sec	per-tenant rate limit returns 429 with Retry-After; limiter follows quota changes.	TestRateLimit	PASS 0.01s
TC-GW-005	Unit	request hygiene: Idempotency-Key format, JSON content type, unknown fields, trailing data, 256 KiB body cap.	TestRequestHygiene	PASS 0.01s
TC-GW-006	Unit	TMF641 order validation: items, ids, actions, catalog, characteristics, CEL/JSON-schema params, dependsOn graph.	TestValidateOrder	PASS 0.01s
TC-GW-007	Unit / Sec	TMF688 callback allow-list (SSRF): https only, exact host:port, no userinfo/fragment, length cap.	TestCallbackAllowed	PASS 0.00s
TC-GW-008	Unit	hub registration input validation happens before any storage access.	TestHubValidation	PASS 0.01s
TC-GW-009	Unit	panics inside handlers are recovered into a TMF630 500 without leaking internals.	TestPanicRecovery	PASS 0.01s
TC-GW-010	Unit	toTMF maps store rows to TMF641 including item states and error messages.	TestToTMF	PASS 0.00s
TC-GW-011	Integration / Sec	create order 201, idempotent replay 200, key reuse with different body 409, get, list.	TestOrderLifecycleIntegration	PASS 0.52s
TC-GW-012	Integration / Sec	tenant isolation through RLS: tenant B cannot read, list or cancel tenant A's order; the same Idempotency-Key in two tenants yields two different orders.	TestTenantIsolationIntegration	PASS 0.45s
TC-GW-013	Integration	host operator acting for a tenant via X-Tenant (dxps:tenant:any) sees that tenant's data only.	TestActingForIntegration	PASS 0.52s
TC-GW-014	Integration / Sec	hub create (generated secret returned once, sealed at rest), delete, cross-tenant delete 404.	TestHubIntegration	PASS 0.42s
TC-GW-015	Integration	storage failures surface as 503 DXPS-9001 (no internal error text).	TestStoreUnavailableIntegration	PASS 0.73s
TC-GW-016	Unit	fast-publish failures are logged and left to the outbox relay.	TestPublishFailure	PASS 0.00s

5.11 ids - Identifiers: UUIDv7 and deterministic derived ids

Design reference: LLD 5 Data model. Source: internal/ids/ids_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
TC-IDS-001	Unit	New identifiers are time-ordered UUIDv7 values.	TestNewV7	PASS 0.00s
TC-IDS-002	Unit	Derived ids are deterministic per tenant + idempotency key and cannot collide by concatenation.	TestDerive	PASS 0.00s
TC-IDS-003	Unit	Idempotency keys and UUID syntax validation.	TestValidation	PASS 0.00s

5.12 keylane - Entity key lanes (per-subscriber ordering)

Design reference: LLD 6.5 Sequencing. Source: internal/keylane/keylane_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
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TC-KLN-001	Unit	Records with the same key run strictly serially in arrival order.	TestSameKeyOrdered	PASS 0.03s
TC-KLN-002	Unit	Only the highest contiguous completed offset is committable (no data loss on crash).	TestContiguousCommit	PASS 0.00s
TC-KLN-003	Unit	A failing (poison) record is reported but does not block its lane; revoked partitions are not committed.	TestPoisonAndRevoke	PASS 0.02s

5.13 netsim - Network element simulators (9 NEs + webhook sink)

Design reference: Test harness. Source: internal/netsim/netsim_test.go. Coverage 99.3%.

ID	Level	Objective and expected result	Go test	Result
TC-SIM-001	Unit / Sec	NRF issues client_credentials tokens and rejects malformed grant requests; discovery answers.	TestNRF	PASS 0.00s
TC-SIM-002	Unit / Sec	UDR rejects missing, forged, expired-scope and wrong-scope bearer tokens (401).	TestUDRAuth	PASS 0.01s
TC-SIM-003	Unit / Sec	UDR auth-subscription validates AKA method and HSM key references; PUT 201 then 204; GET; DELETE.	TestUDRAuthSubscription	PASS 0.00s
TC-SIM-004	Unit	shared UDR isolates tenants: another tenant cannot overwrite (409) or delete (403) a subscriber.	TestUDRTenantIsolation	PASS 0.00s
TC-SIM-005	Unit	UDR PATCH applies RFC 7396 merge patch and requires application/merge-patch+json.	TestUDRMergePatch	PASS 0.00s
TC-SIM-006	Unit	other UDR resources (SMF selection, policy am-data) accept PUT without extra validation.	TestUDROtherResources	PASS 0.00s
TC-SIM-007	Unit	IMS-PG enforces IMPI -> IMPU -> MMTEL ordering (409 PRECONDITION) and tenant ownership.	TestIMS	PASS 0.00s
TC-SIM-008	Unit	RESTCONF YANG-Patch (RFC 8072): media type, structure, capacity check (409 yang-patch-status).	TestRESTCONF	PASS 0.00s
TC-SIM-009	Unit	USP (TR-369) Set returns SET_RESP; offline agents report errCode 7002; bad messages 400.	TestUSP	PASS 0.00s
TC-SIM-010	Unit	SM-DP+ ES2+ download/confirm/cancel flow, protocol header and business errors in 200 responses.	TestSMDP	PASS 0.00s
TC-SIM-011	Unit / Sec	OCS/BSS TMF666/637: create, X-Request-ID replay, ALREADY_EXISTS, missing msisdn, brand isolation.	TestOCS	PASS 0.00s
TC-SIM-012	Unit	NPDB port-in requires body msisdn == path msisdn; delete.	TestNPDB	PASS 0.00s
TC-SIM-013	Unit	NEF CAMARA QoD session create requires E.164 phone and qosProfile; delete by msisdn.	TestNEF	PASS 0.00s
TC-SIM-014	Unit / Sec	webhook sink records TMF688 deliveries with tenant, event id and signature (ordered by time).	TestSink	PASS 0.01s
TC-SIM-015	Unit / Sec	fault injection by key suffix: 999 -> 503 + Retry-After, 998 -> 400, 997 -> two 503s then success.	TestFaultInjection	PASS 0.00s
TC-SIM-016	Unit	configured random error rate makes every call fail with 503 when set to 1.0.	TestRandomErrors	PASS 0.00s
TC-SIM-017	Unit	request bodies above 256 KiB are rejected with 413 before reaching the simulator.	TestBodyLimit	PASS 0.00s
TC-SIM-018	Unit	exchange log captures compacted request/response payloads, tenant, status; bounded to 300.	TestExchangeLog	PASS 0.01s
TC-SIM-019	Unit / Sec	admin API requires the HMAC admin token; validates fault requests; applies latency/error rate.	TestAdmin	PASS 0.00s
TC-SIM-020	Unit / Sec	admin listener refuses non-loopback binds and serves on loopback.	TestServeAdmin	PASS 0.00s
TC-SIM-021	Unit / Sec	simulators serve HTTP/2 over mutual TLS; clients without a certificate are rejected.	TestServeMTLS	PASS 0.02s
TC-SIM-022	Security	the exchange log (shown on the dashboard, exported to reports) never carries credentials: NRF access tokens, nested secrets and form-encoded secrets are masked; other fields are kept.	TestExchangeRedaction	PASS 0.00s

5.14 orchestrator - Orchestrator: sagas, dependencies, sequencing (E2E)

Design reference: LLD 6.3-6.6. Source: internal/orchestrator/orchestrator_test.go. Coverage 78.2%.

ID	Level	Objective and expected result	Go test	Result
TC-ORC-001	Unit	DXPS error codes are extracted from wrapped errors; default otherwise.	TestErrCode	PASS 0.00s
TC-ORC-002	Unit	transient errors are retried with backoff; permanent errors and cancelled contexts stop at once.	TestWithRetry	PASS 0.15s
TC-ORC-003	Unit / Sec	the stats endpoint refuses non-loopback binds.	TestServeStatsLoopback	PASS 0.00s
TC-ORC-004	E2E / Sec	CreateSubscriber5G runs its DAG (UDR auth -> am/smf/policy -> OCS) to completion; TMF688 events report acknowledged -> inProgress -> completed.	TestCreateSubscriberE2E	PASS 1.03s
TC-ORC-005	E2E	a permanent NE rejection (998) in an ATOMIC command compensates every succeeded task in reverse order, fails the order and writes an audit record to the DLQ.	TestCompensationE2E	PASS 1.05s
TC-ORC-006	E2E	transient 503s (997) are retried through the retry topics and the order	TestRetryLadderE2E	PASS 1.07s

		completes.		
TC-ORC-007	E2E	retries are bounded: a permanently unavailable NE (999) exhausts MaxAttempts -> DXPS-4001, rollback.	TestRetryExhaustedE2E	PASS 1.34s
TC-ORC-008	E2E	order item dependsOn: AddVoNR waits for CreateSubscriber5G, then runs IMS IMPI -> IMPU -> MMTEL.	TestItemDependencyE2E	PASS 1.39s
TC-ORC-008b	E2E	a failed prerequisite item fails the dependent item without calling its NES.	TestItemDependencyFailedE2E	PASS 1.17s
TC-ORC-009	E2E	two orders on the same subscriber are serialised: the second is parked until the first finishes, then admitted automatically.	TestEntitySequencerE2E	PASS 1.22s
TC-ORC-010	Unit	BCs for unknown specs or suspended tenants are rejected (order state rejected); duplicates are ignored; tenant header/key mismatches and invalid messages are permanent errors (DLQ).	TestHandleBCGuards	PASS 0.54s
TC-ORC-011	Unit	TaskResult guards: tenant mismatch / invalid ids are permanent; unknown orders, stale attempts and duplicate results are ignored.	TestHandleResultGuards	PASS 0.78s
TC-ORC-012	Integration / Sec	the service wiring starts the tiered BC/result consumers, outbox relay, retry forwarder, registry refresh and the loopback stats endpoint, then shuts down cleanly on cancel.	TestServiceRun	PASS 1.75s

5.15 pki - PKI and mutual TLS

Design reference: LLD 7.2 Transport security. Source: internal/pki/pki_test.go. Coverage 89.1%.

ID	Level	Objective and expected result	Go test	Result
TC-PKI-001	Unit / Sec	Mutual TLS 1.3 over HTTP/2 succeeds with a client certificate from the DxPS CA.	TestMutualTLS	PASS 0.02s
TC-PKI-002	Unit	The NE side rejects clients without a certificate or with a certificate from a foreign CA.	TestMTLSRejections	PASS 0.02s
TC-PKI-003	Unit / Sec	Clients refuse servers not signed by the DxPS CA and TLS < 1.3.	TestServerVerification	PASS 0.01s
TC-PKI-004	Unit	Leaf certificates carry correct key usages, SANs and validity; CA files round-trip with private key 0600.	TestBundleFiles	PASS 0.18s

5.16 planner - Planner: business command to NE task DAG

Design reference: LLD 6.3 Planning. Source: internal/planner/planner_test.go. Coverage 96.2%.

ID	Level	Objective and expected result	Go test	Result
TC-PLN-001	Unit	CreateSubscriber5G decomposes into the UDR/OCS DAG; only roots are rendered at plan time.	TestCreateSubscriberDAG	PASS 0.03s
TC-PLN-002	Unit	A CEL 'when' that is false skips the task; dependants inherit its dependencies.	TestWhenSkip	PASS 0.02s
TC-PLN-003	Unit	Tenant override spec and own NE are used for a FULL MVNO (mvno-alpha own OCS + welcome bundle).	TestTenantOverride	PASS 0.02s
TC-PLN-004	Unit	Planning fails with DXPS-1005 when the tenant is not entitled to the required NE.	TestNotEntitled	PASS 0.01s
TC-PLN-005	Unit	Tasks on the same resource are coalesced into a not-yet-dispatched task of the same order.	TestCoalesce	PASS 0.01s
TC-PLN-006	Unit	Late rendering - eSIM ConfirmOrder uses the ICCID returned by DownloadOrder; compensation renders CancelOrder.	TestLateRendering	PASS 0.02s
TC-PLN-007	Unit	A spec whose CEL expressions fail at evaluation time is rejected with ErrParams.	TestCELRuntimeErrors	PASS 0.01s

5.17 registry - NE registry, shared-NE access and quotas

Design reference: LLD 6.7 NE registry. Source: internal/registry/registry_test.go. Coverage 98.8%.

ID	Level	Objective and expected result	Go test	Result
TC-REG-001	Unit	A FULL MVNO uses its own NE first (own OCS) and falls back to entitled shared NEs.	TestSelectOwnFirst	PASS 0.00s
TC-REG-002	Unit	Tenants without an entitlement get DXPS-1005; unknown NF types DXPS-1003.	TestEntitlement	PASS 0.00s
TC-REG-003	Unit	Endpoints are ordered by weight; NEs without endpoints are unresolvable.	TestEndpoint	PASS 0.00s
TC-REG-004	Unit	The NE concurrency cap blocks extra callers until a slot is released.	TestConcurrencyCap	PASS 0.03s
TC-REG-005	Unit	An open circuit breaker rejects calls to a failing NE.	TestBreakerGate	PASS 0.00s
TC-REG-006	Unit	Per-tenant quota on a shared NE throttles a noisy MVNO without affecting others.	TestTenantQuota	PASS 0.10s

5.18 saga - Saga compensation and retry back-off

Design reference: LLD 6.6 Resilience. Source: internal/saga/saga_test.go. Coverage 98.8%.

ID	Level	Objective and expected result	Go test	Result
TC-SAG-001	Unit	NE responses are classified: 2xx success, 408/429/5xx(gateway) retryable, other 4xx/501 failed, transport errors retryable.	TestClassify	PASS 0.00s
TC-SAG-002	Unit	Backoff is exponential with full jitter, floored at 50 ms and capped at 5 min; Retry-After wins when larger.	TestBackoff	PASS 0.00s
TC-SAG-003	Unit	Retries stop at the attempt budget or the order deadline.	TestExhausted	PASS 0.00s
TC-SAG-004	Unit	Compensation runs in reverse topological order (dependants first), skipping failed/non-compensable tasks; nothing is compensated after a succeeded pivot.	TestCompensationOrder	PASS 0.00s
TC-SAG-005	Unit	TMF641 order state is derived from the states of its business commands.	TestOrderState	PASS 0.00s

5.19 scheduler - Deficit round-robin fair scheduler (tenant x priority)

Design reference: LLD 6.4 Fairness. Source: internal/scheduler/scheduler_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
TC-SCH-001	Unit	P0 (fraud/lawful barring) is always served before any queued P1..P3 work.	TestStrictP0	PASS 0.00s
TC-SCH-002	Unit	P1:P2:P3 are served in weighted round robin 4:2:1 under saturation.	TestWeightedRoundRobin	PASS 0.00s
TC-SCH-003	Unit	Inside a tier, tenants share by deficit round robin in proportion to their SLA weight; a noisy tenant cannot starve a small MVNO.	TestTenantDRR	PASS 0.00s
TC-SCH-004	Unit	Bounded capacity applies backpressure; Push unblocks when space frees and honours ctx.	TestBackpressure	PASS 0.07s
TC-SCH-005	Unit	Next blocks until an item arrives and returns ctx errors when cancelled.	TestNextBlocking	PASS 0.02s

5.20 secretbox - Secret encryption at rest (AES-256-GCM)

Design reference: LLD 7.3 Secrets. Source: internal/secretbox/secretbox_test.go. Coverage 87.5%.

ID	Level	Objective and expected result	Go test	Result
TC-SBX-001	Unit / Sec	Webhook secrets round-trip through AES-256-GCM and are never stored in clear.	TestRoundTrip	PASS 0.00s
TC-SBX-002	Unit	A ciphertext is bound to its tenant (AAD) - it cannot be opened under another tenant.	TestTenantBinding	PASS 0.00s
TC-SBX-003	Unit / Sec	Tampered, truncated or foreign-key ciphertexts and bad keys are rejected.	TestTamper	PASS 0.00s

5.21 store - PostgreSQL store: RLS, outbox, migrations, seed

Design reference: LLD 5 Data model. Source: internal/store/store_test.go. Coverage 81.5%.

ID	Level	Objective and expected result	Go test	Result
TC-STO-001	Unit	outbox shard is stable and within range; error classifiers.	TestShardAndErrors	PASS 0.00s
TC-STO-002	Integration / Sec	a tenant transaction sees only its own rows even with an explicit WHERE on another tenant, cannot insert rows for another tenant (WITH CHECK), and cannot delete another tenant's rows.	TestRowLevelSecurity	PASS 0.30s
TC-STO-003	Integration / Sec	without the tenant setting the app role sees nothing; invalid tenant ids are refused before any SQL runs.	TestNoTenantContext	PASS 0.23s
TC-STO-004	Integration / Sec	least privilege: the app role cannot update/delete the append-only audit log or touch tables it has no grant on.	TestLeastPrivilege	PASS 0.27s
TC-STO-005	Integration / Sec	outbox rows get ids; MarkPublished is tenant-scoped and idempotent; relay propagates publish failures without marking rows.	TestOutbox	PASS 0.22s
TC-STO-006	Integration	registry and tenant loaders, statistics for the dashboard.	TestOpsLoaders	PASS 0.87s
TC-STO-007	Integration	migrations are versioned and idempotent; a re-run applies nothing and a bad DSN fails cleanly.	TestMigrateIdempotent	PASS 0.32s
TC-STO-008	Integration	re-applying the seed (tenants, NEs + endpoints, shared-NE access, catalog specs) is idempotent: same tenants, same NE ids, same access grants. Seed functions need the ops pool.	TestSeedIdempotent	PASS 0.33s

5.22 svc - Service bootstrap, logging, HTTP server hardening

Design reference: LLD 8 Deployment. Source: internal/svc/svc_test.go. Coverage 75.9%.

ID	Level	Objective and expected result	Go test	Result
TC-SVC-001	Unit	Boot loads configuration, writes JSON logs to \$DXPS_LOG_DIR/<name>.log (0600) and returns a cancellable context.	TestBoot	PASS 0.04s
TC-SVC-002	Unit	HTTP servers get conservative timeouts and header limits (slowloris protection).	TestServer	PASS 0.00s
TC-SVC-003	Integration	OpenStore connects both pools.	TestOpenStore	PASS 0.25s

5.23 tenant - Tenant registry, identifiers and SLA weights

Design reference: LLD 4 Multi-tenancy. Source: internal/tenant/tenant_test.go. Coverage 100.0%.

ID	Level	Objective and expected result	Go test	Result
TC-TEN-001	Unit	Tenant identifiers follow the dxps_tenant domain (1..40 safe characters).	TestValid	PASS 0.00s
TC-TEN-002	Unit	Require enforces valid, known and ACTIVE tenants (DXPS-1004).	TestRequire	PASS 0.00s
TC-TEN-003	Unit	Registry replace/all/quantum are consistent and concurrency safe.	TestRegistry	PASS 0.00s
TC-TEN-004	Unit	Tenant context propagation.	TestContext	PASS 0.00s

6. Security verification

6.1 Live attack probes (running stack)

The mosaic's Run probes action attacks the live gateway and an NE simulator with 24 probes. Results below are from the final rebuilt stack. The same probes run in TC-DSH-010 against a test stack.

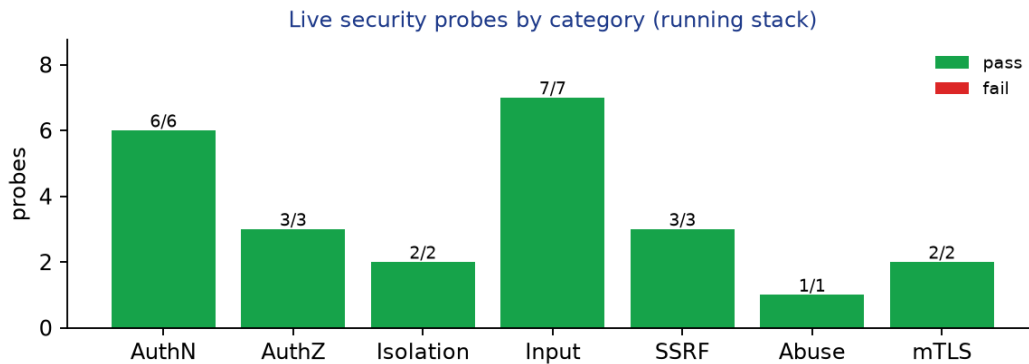


Figure 2 - Live security probes by category

ID	Category	Attack	Expected	Observed	Result
SEC-01	AuthN	No bearer token	401	401 DXPS-1000	PASS
SEC-02	AuthN	JWT alg=none	401	401 DXPS-1000	PASS
SEC-03	AuthN	Tampered JWT signature	401	401 DXPS-1000	PASS
SEC-04	AuthN	Tenant claim swapped (signature reused)	401	401 DXPS-1000	PASS
SEC-05	AuthN	Expired JWT	401	401 DXPS-1000	PASS
SEC-06	AuthN	Wrong audience	401	401 DXPS-1000	PASS
SEC-07	AuthZ	X-Tenant impersonation without dxps:tenant:any	403 DXPS-1004	403 DXPS-1004	PASS
SEC-08	AuthZ	Write with read-only scope	403	403 DXPS-1000	PASS
SEC-09	AuthZ	Suspended tenant	403 DXPS-1004	403 DXPS-1004	PASS
SEC-10	Isolation	Cross-tenant read of another tenant's order (RLS)	404	404 DXPS-1404	PASS
SEC-11	Isolation	Owner can read the same order	200	200	PASS
SEC-12	Input	Unknown JSON field	400 DXPS-1002	400 DXPS-1002	PASS
SEC-13	Input	Trailing JSON (request smuggling)	400 DXPS-1002	400 DXPS-1002	PASS
SEC-14	Input	Oversized body (300 KiB)	413 DXPS-1002	413 DXPS-1002	PASS
SEC-15	Input	SQL injection in characteristic	400 DXPS-1002	400 DXPS-1002	PASS
SEC-16	Input	SQL injection in path id	404	404 DXPS-1404	PASS
SEC-17	Input	Missing Idempotency-Key	400	400 DXPS-1002	PASS
SEC-18	Input	Idempotency-Key reused with different body	409 DXPS-1008	409 DXPS-1008	PASS
SEC-19	SSRF	Webhook to cloud metadata address	400	400 DXPS-1002	PASS
SEC-20	SSRF	Webhook over plain http	400	400 DXPS-1002	PASS
SEC-21	SSRF	Webhook with userinfo	400	400 DXPS-1002	PASS
SEC-22	Abuse	Per-tenant rate limit (80 parallel calls)	some 429 DXPS-1006	59 x 429	PASS
SEC-23	mTLS	NE simulator without client certificate	handshake rejected	rejected: certificate required	PASS
SEC-24	mTLS	NE simulator with certificate from a foreign CA	handshake rejected	rejected: unknown certificate authority	PASS

6.2 Security controls verified by automated tests

Control	Verified by
JWT: HS256 only; alg=none / RS256 / HS512 / wrong typ rejected; expiry, issuer, audience, tenant format	TC-AUTH-001..009, SEC-01..06
Scope-based authorisation (order read/write, hub write)	TC-GW, SEC-07..09
Tenant isolation: PostgreSQL row-level security on every table, WITH CHECK on insert, no rows without tenant context	TC-STO-002, TC-STO-003, SEC-10/11
Least privilege: app role cannot UPDATE/DELETE audit, TRUNCATE, DROP, ALTER or disable RLS	TC-STO-004
Input validation: schema, patterns, enum, size limits (413), unknown fields, JSON injection into NE templates	TC-CAT-003/004, TC-GW, SEC-12..18
Missing dependency data fails the render instead of sending null to an NE	TC-CAT-007
SSRF: webhook callbacks must be https and on an explicit host:port allow-list (checked at registration and again at delivery); redirects are never followed	TC-GW, TC-HUB, SEC-19..21
Abuse: per-tenant token-bucket rate limit (429 DXPS-1006)	TC-GW, SEC-22
Mutual TLS to every NE; certificates from a foreign CA or missing certificates refused	TC-SIM-021, TC-ADP, TC-DSH-013, SEC-23/24
Webhook integrity: HMAC-SHA256 signatures, secrets encrypted at rest (AES-256-GCM)	TC-HUB, TC-SBX
Dashboard: loopback bind, Host allow-list (DNS rebinding), Origin + double-submit CSRF, security headers	TC-DSH-001..004
No secrets in logs or exchange records (tokens, client secrets, passwords redacted)	TC-SVC-001, TC-SIM-022, TC-CFG-003
Idempotency: derived order ids, replay returns 200 + Idempotent-Replayed, key reuse with a different body 409	TC-GW, SEC-18

6.3 Static analysis

Tool	Result
govulncheck (golang.org/x/vuln v1.8.0)	No vulnerabilities found
gosec v2.29.0	45 files, 10,211 lines: 0 high, 0 medium, 20 low; 9 reviewed #nosec annotations
go vet	clean

- **First govulncheck run:** 7 reachable vulnerabilities, 6 in the Go 1.26.5 standard library (net/url, crypto/tls, net/http, encoding/xml, encoding/asn1, idna) and 1 in golang.org/x/text v0.29.0. Fixed by moving to toolchain go1.26.6 and x/text v0.42.0; the re-scan is clean.
- **gosec G404 (math/rand) excluded:** math/rand is used only for retry jitter, simulator latency and the demo traffic mix. Tokens, CSRF nonces, certificate serials and keys use crypto/rand.
- **Reviewed #nosec:** file paths that come from operator configuration (runtime directory, PKI directory), never from requests (G304/G703); public certificate files written 0644 while keys are 0600 (G306); the CSRF cookie's Secure flag follows the transport because the mosaic is plain HTTP on loopback (G124).
- **Remaining low findings (G104):** 20 ignored errors on response writes, Body.Close and flag.Parse(ExitOnError); reviewed as benign.

7. Defects found and fixed during testing

#	Defect	Impact	Fix	Regression test
1	Simulator exchange log stored NRF OAuth access tokens; the log is shown on the dashboard	Credential exposure (CWE-532)	Recursive redaction of token/secret/password fields in JSON and form bodies	TC-SIM-022
2	Dashboard accepted any Host header	DNS rebinding could drive the loopback dashboard from a web page	Host allow-list, 421 Misdirected Request otherwise	TC-DSH-002
3	mTLS probe counted an unreachable simulator as 'rejected'; then misread a Windows TCP reset	False security pass / flaky probe	Phase-based verdict: TCP connect, handshake, first read; timeouts and server-certificate errors never pass	TC-DSH-010, TC-DSH-013
4	Event hub retried HTTP 3xx responses	Redirect-driven request amplification toward other hosts	Only 408, 429 and 5xx are retried; redirects are never followed	TC-HUB
5	A missing dependency result (eSIM iccid) rendered as JSON null into the NE payload	Wrong data sent to a network element	Template function need fails the render -> permanent task error	TC-CAT-007
6	Go 1.26.5 standard library and x/text CVEs reachable	DoS / parsing issues on TLS and HTTP paths	go1.26.6, x/text v0.42.0	govulncheck
7	Env-file loader ignored os.Setenv errors	A bad secrets line silently	Error naming the key (never the value)	TC-CFG-003

		dropped configuration		
8	Kafka on Windows: log cleaner / retention rename of open segments took the log dir offline	Broker shutdown in the dev runtime	Dev runtime disables the log cleaner and retention checks; topics are never deleted by tests (production on Linux keeps defaults)	Stack stable through all runs

8. End-to-end and load results

Mosaic load run: 2000 orders, 100 orders/s, chaos 0.1 (NE fault suffixes 997/998/999), all 6 services + Kafka 4.3.1 + PostgreSQL 18.6 on one Windows laptop.

2,000	2,000	115 ms	2335 ms	0
orders sent	accepted (201)	gateway p50	gateway p99	outbox backlog after

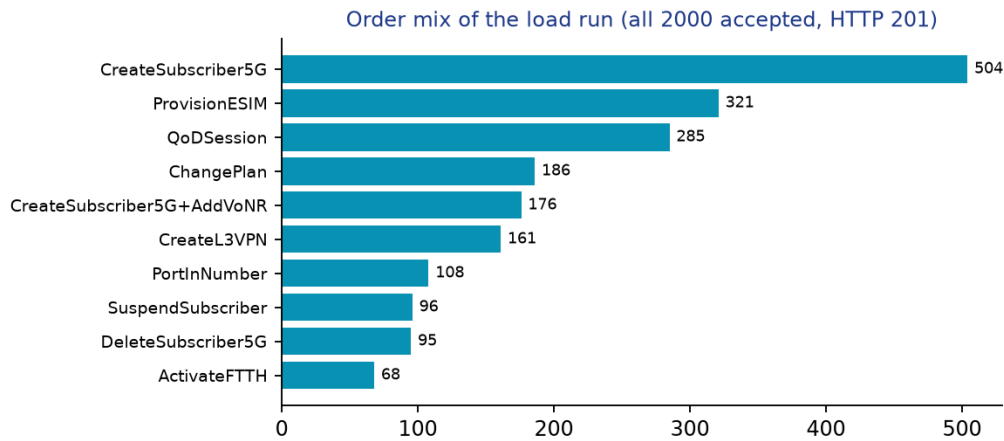


Figure 3 - Order mix of the load run

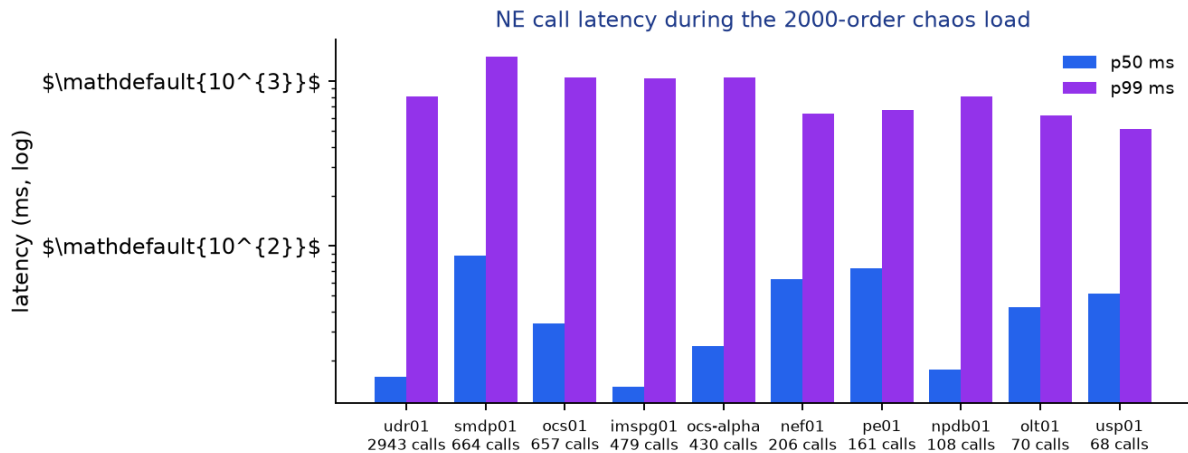


Figure 4 - NE call latency (log scale); error counts include the injected chaos faults

NE	Calls	p50 ms	p99 ms	Errors (incl. injected)
udr01	2,943	16.0	804	128
smdp01	664	87.6	1416	46
ocs01	657	33.6	1051	22
imspg01	479	13.9	1040	6
ocs-alpha	430	24.8	1049	15
nef01	206	62.7	638	0
pe01	161	72.7	667	11

npdb01	108	17.8	810	0
olt01	70	42.2	623	0
usp01	68	51.2	513	2

Interpretation

- Every order was accepted and persisted (2000 x HTTP 201). The transactional outbox drained to zero, so no command was lost between PostgreSQL and Kafka.
- NE errors match the chaos settings: 997 suffixes recover via the retry ladder; 998/999 fail and compensate.
- Latencies are from one laptop running Kafka, PostgreSQL, six services, the simulators and the load generator at once. They are functional evidence, not a capacity benchmark. The p99 order time (about 18 s) includes orders that deliberately walked the retry ladder.

Appendix A - End-to-end payload traces

Captured by the orchestrator E2E tests with DXPS_CAPTURE_DIR set. Each trace shows the real TMF641 request and responses from the gateway, the TMF688 state changes, and every NE call the adapter made to the simulators, in order.

A.1 CreateSubscriber5G - happy path (TC-ORC-004)

Tenant: mvno-beta **TMF688 states:** acknowledged -> inProgress -> completed

TMF641 POST /serviceOrder (request)

```
{
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278901"
          },
          {
            "name": "msisdn",
            "value": "141686278901"
          }
        ]
      }
    }
  ]
}
```

201 Created (response)

```
{
  "@type": "ServiceOrder",
  "id": "fa0788c4-9bf5-5535-bda0-76dd7706f6ab",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/fa0788c4-9bf5-5535-bda0-76dd7706f6ab",
  "priority": "1",
  "state": "acknowledged",
  "orderDate": "2026-10-06T20:43:16.341914-07:00",
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278901"
          },
          {
            "name": "msisdn",
            "value": "141686278901"
          }
        ]
      }
    }
  ]
}
```

```

    },
    "state": "acknowledged"
  }
]
}

```

NE calls (6)

#	Simulator	Method	Path	Status	Fault
1	nrf	POST	/oauth2/token	200	
2	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278901/authentication-data/authentication-subscription	201	
3	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278901/41677/provisioned-data/am-data	201	
4	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278901/41677/provisioned-data/smf-selection-subscription-data	201	
5	udr	PUT	/nudr-dr/v2/policy-data/ues/imsi-416775586278901/am-data	201	
6	ocs-bss	POST	/tmf-api/accountManagement/v5/billingAccount	201	

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278901/authentication-data/authentication-subscription -> 201

```

request:
{
  "algorithmId": "milenaage",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpcKey": "hsm://mvno-beta/opc/imsi-416775586278901",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278901",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "000000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278901"
}
response:
{
  "algorithmId": "milenaage",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpcKey": "hsm://mvno-beta/opc/imsi-416775586278901",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278901",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "000000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278901"
}

```

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278901/41677/provisioned-data/am-data -> 201

```

request:
{
  "gpsis": [
    "msisdn-141686278901"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ]
}

```

```

    ],
    "subscribedUeAmbr": {
      "downlink": "1 Gbps",
      "uplink": "200 Mbps"
    },
    "vendorSpecific-dxps": {
      "plan": "5G-100GB",
      "tenant": "mvno-beta"
    }
  }
}
response:
{
  "gpsis": [
    "msisdn-141686278901"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
  "subscribedUeAmbr": {
    "downlink": "1 Gbps",
    "uplink": "200 Mbps"
  },
  "vendorSpecific-dxps": {
    "plan": "5G-100GB",
    "tenant": "mvno-beta"
  }
}

```

Final order (GET /serviceOrder/{id})

```

{
  "@type": "ServiceOrder",
  "id": "fa0788c4-9bf5-5535-bda0-76dd7706f6ab",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/fa0788c4-9bf5-5535-bda0-76dd7706f6ab",
  "priority": "1",
  "state": "completed",
  "orderDate": "2026-10-06T20:43:16.341914-07:00",
  "completionDate": "2026-10-06T20:43:16.738659-07:00",
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278901"
          },
          {
            "name": "msisdn",
            "value": "141686278901"
          }
        ]
      },
      "state": "completed"
    }
  ]
}

```

```
]
}
```

A.2 CreateSubscriber5G + AddVoNR with dependsOn (TC-ORC-008)

Tenant: mvno-beta **TMF688 states:** acknowledged -> inProgress -> completed

TMF641 POST /serviceOrder (request)

```
{
  "serviceOrderItem": [
    {
      "id": "sub",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278905"
          },
          {
            "name": "msisdn",
            "value": "141686278905"
          }
        ]
      }
    }
  ],
  {
    "id": "volte",
    "action": "add",
    "service": {
      "serviceSpecification": {
        "id": "AddVoNR"
      },
      "serviceCharacteristic": [
        {
          "name": "supi",
          "value": "imsi-416775586278905"
        },
        {
          "name": "msisdn",
          "value": "141686278905"
        }
      ]
    }
  },
  "serviceOrderItemRelationship": [
    {
      "relationshipType": "dependsOn",
      "orderItem": {
        "itemId": "sub"
      }
    }
  ]
}
```

201 Created (response)

```
{
  "@type": "ServiceOrder",
  "id": "d9e8f3cb-2cf1-5f23-8429-8781c9eeb0d1",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/d9e8f3cb-2cf1-5f23-8429-8781c9eeb0d1",
  "priority": "1",
  "state": "acknowledged",
  "orderDate": "2026-10-06T20:43:19.514855-07:00",
}
```

```

"serviceOrderItem": [
  {
    "id": "sub",
    "action": "add",
    "service": {
      "serviceSpecification": {
        "id": "CreateSubscriber5G"
      },
      "serviceCharacteristic": [
        {
          "name": "plan",
          "value": "5G-100GB"
        },
        {
          "name": "supi",
          "value": "imsi-416775586278905"
        },
        {
          "name": "msisdn",
          "value": "141686278905"
        }
      ]
    },
    "state": "acknowledged"
  },
  {
    "id": "volte",
    "action": "add",
    "service": {
      "serviceSpecification": {
        "id": "AddVoNR"
      },
      "serviceCharacteristic": [
        {
          "name": "supi",
          "value": "imsi-416775586278905"
        },
        {
          "name": "msisdn",
          "value": "141686278905"
        }
      ]
    },
    "state": "acknowledged"
  },
  ... (truncated)

```

NE calls (9)

#	Simulator	Method	Path	Status	Fault
1	nrf	POST	/oauth2/token	200	
2	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278905/authentication-data/authentication-subscription	201	
3	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278905/41677/provisioned-data/am-data	201	
4	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278905/41677/provisioned-data/smf-selection-subscription-data	201	
5	udr	PUT	/nudr-dr/v2/policy-data/ues/imsi-416775586278905/am-data	201	
6	ocs-bss	POST	/tmf-api/accountManagement/v5/billingAccount	201	
7	ims-pg	PUT	/ims-pg/v1/subscribers/imsi-416775586278905/impi	200	
8	ims-pg	PUT	/ims-pg/v1/subscribers/imsi-416775586278905/impu	200	
9	ims-pg	PUT	/ims-pg/v1/subscribers/imsi-416775586278905/mmtel	200	

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278905/authentication-data/authentication-subscription -> 201

```

request:
{
  "algorithmId": "milena",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpKey": "hsm://mvno-beta/opc/imsi-416775586278905",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278905",
  "protectionParameterId": "pp-01",

```

```

"sequenceNumber": {
  "lastIndexes": {
    "ausf": 0
  },
  "sqn": "000000000020",
  "sqnScheme": "NON_TIME_BASED"
},
"supi": "imsi-416775586278905"
}
response:
{
  "algorithmId": "milenage",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpcKey": "hsm://mvno-beta/opc/imsi-416775586278905",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278905",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "000000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278905"
}

```

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278905/41677/provisioned-data/am-data -> 201

```

request:
{
  "gpsis": [
    "msisdn-141686278905"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
  "subscribedUeAmbr": {
    "downlink": "1 Gbps",
    "uplink": "200 Mbps"
  },
  "vendorSpecific-dxps": {
    "plan": "5G-100GB",
    "tenant": "mvno-beta"
  }
}
response:
{
  "gpsis": [
    "msisdn-141686278905"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
}

```

```
"subscribedUeAmbr": {
  "downlink": "1 Gbps",
  "uplink": "200 Mbps"
},
"vendorSpecific-dxps": {
  "plan": "5G-100GB",
  "tenant": "mvno-beta"
}
}
```

Final order (GET /serviceOrder/{id})

```
{
  "@type": "ServiceOrder",
  "id": "d9e8f3cb-2cf1-5f23-8429-8781c9eeb0d1",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/d9e8f3cb-2cf1-5f23-8429-8781c9eeb0d1",
  "priority": "1",
  "state": "completed",
  "orderDate": "2026-10-06T20:43:19.514855-07:00",
  "completionDate": "2026-10-06T20:43:20.149949-07:00",
  "serviceOrderItem": [
    {
      "id": "sub",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278905"
          },
          {
            "name": "msisdn",
            "value": "141686278905"
          }
        ]
      },
      "state": "completed"
    },
    {
      "id": "volte",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "AddVoNR"
        },
        "serviceCharacteristic": [
          {
            "name": "supi",
            "value": "imsi-416775586278905"
          },
          {
            "name": "msisdn",
            "value": "141686278905"
          }
        ]
      },
      "serviceOrderItemRelationship": [
        {
          "relationshipType": "dependsOn",
          "orderItem": {
            "itemId": "sub"
          }
        }
      ],
      "state": "completed"
    }
  ]
}
```

```
]
}
```

A.3 Transient NE failure recovered by the retry ladder (TC-ORC-006)

Tenant: mvno-beta **TMF688 states:** acknowledged -> inProgress -> completed

TMF641 POST /serviceOrder (request)

```
{
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278903"
          },
          {
            "name": "msisdn",
            "value": "141686278997"
          }
        ]
      }
    }
  ]
}
```

201 Created (response)

```
{
  "@type": "ServiceOrder",
  "id": "739af6c0-a02b-53fe-b3ee-86241683a9b9",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/739af6c0-a02b-53fe-b3ee-86241683a9b9",
  "priority": "1",
  "state": "acknowledged",
  "orderDate": "2026-10-06T20:43:17.93269-07:00",
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278903"
          },
          {
            "name": "msisdn",
            "value": "141686278997"
          }
        ]
      }
    }
  ],
  "state": "acknowledged"
}
```

NE calls (10)

#	Simulator	Method	Path	Status	Fault
1	nrf	POST	/oauth2/token	200	
2	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278903/authentication-data/authentication-subscription	201	
3	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278903/41677/provisioned-data/am-data	503	flaky
4	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278903/41677/provisioned-data/smf-selection-subscription-data	201	
5	udr	PUT	/nudr-dr/v2/policy-data/ues/imsi-416775586278903/am-data	201	
6	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278903/41677/provisioned-data/am-data	503	flaky
7	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278903/41677/provisioned-data/am-data	201	flaky-recovered
8	ocs-bss	POST	/tmf-api/accountManagement/v5/billingAccount	503	flaky
9	ocs-bss	POST	/tmf-api/accountManagement/v5/billingAccount	503	flaky
10	ocs-bss	POST	/tmf-api/accountManagement/v5/billingAccount	201	flaky-recovered

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278903/authentication-data/authentication-subscription -> 201

```
request:
{
  "algorithmId": "milena",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpKey": "hsm://mvno-beta/opc/imsi-416775586278903",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278903",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "00000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278903"
}
response:
{
  "algorithmId": "milena",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpKey": "hsm://mvno-beta/opc/imsi-416775586278903",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278903",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "00000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278903"
}
```

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278903/41677/provisioned-data/am-data -> 503

```
request:
{
  "gpsis": [
    "msisdn-141686278997"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
  "subscribedUeAmbr": {
    "downlink": "1 Gbps",
```

```

    "uplink": "200 Mbps"
  },
  "vendorSpecific-dxps": {
    "plan": "5G-100GB",
    "tenant": "mvno-beta"
  }
}
response:
{
  "cause": "TEMPORARILY_UNAVAILABLE",
  "detail": "simulated transient failure",
  "status": 503,
  "title": "Service Unavailable"
}

```

Final order (GET /serviceOrder/{id})

```

{
  "@type": "ServiceOrder",
  "id": "739af6c0-a02b-53fe-b3ee-86241683a9b9",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/739af6c0-a02b-53fe-b3ee-86241683a9b9",
  "priority": "1",
  "state": "completed",
  "orderDate": "2026-10-06T20:43:17.93269-07:00",
  "completionDate": "2026-10-06T20:43:18.299503-07:00",
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278903"
          },
          {
            "name": "msisdn",
            "value": "141686278997"
          }
        ]
      },
      "state": "completed"
    }
  ]
}

```

A.4 Permanent NE rejection -> ATOMIC saga compensation (TC-ORC-005)

Tenant: mvno-beta **TMF688 states:** acknowledged -> inProgress -> failed

TMF641 POST /serviceOrder (request)

```

{
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {

```

```

      "name": "supi",
      "value": "imsi-416775586278902"
    },
    {
      "name": "msisdn",
      "value": "141686278998"
    }
  ]
}
]
}
}

```

201 Created (response)

```

{
  "@type": "ServiceOrder",
  "id": "44a48169-3aae-50a2-9e8a-455f61bc205d",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/44a48169-3aae-50a2-9e8a-455f61bc205d",
  "priority": "1",
  "state": "acknowledged",
  "orderDate": "2026-10-06T20:43:17.163208-07:00",
  "serviceOrderItem": [
    {
      "id": "1",
      "action": "add",
      "service": {
        "serviceSpecification": {
          "id": "CreateSubscriber5G"
        },
        "serviceCharacteristic": [
          {
            "name": "plan",
            "value": "5G-100GB"
          },
          {
            "name": "supi",
            "value": "imsi-416775586278902"
          },
          {
            "name": "msisdn",
            "value": "141686278998"
          }
        ]
      },
      "state": "acknowledged"
    }
  ]
}

```

NE calls (8)

#	Simulator	Method	Path	Status	Fault
1	nrf	POST	/oauth2/token	200	
2	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278902/authentication-data/authentication-subscription	201	
3	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278902/41677/provisioned-data/am-data	400	invalid
4	udr	PUT	/nudr-dr/v2/subscription-data/imsi-416775586278902/41677/provisioned-data/smf-selection-subscription-data	201	
5	udr	PUT	/nudr-dr/v2/policy-data/ues/imsi-416775586278902/am-data	201	
6	udr	DELETE	/nudr-dr/v2/policy-data/ues/imsi-416775586278902/am-data	204	
7	udr	DELETE	/nudr-dr/v2/subscription-data/imsi-416775586278902/41677/provisioned-data/smf-selection-subscription-data	204	
8	udr	DELETE	/nudr-dr/v2/subscription-data/imsi-416775586278902/authentication-data/authentication-subscription	204	

PUT udr /nudr-dr/v2/subscription-data/imsi-416775586278902/authentication-data/authentication-subscription -> 201

```

request:
{
  "algorithmId": "milena",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",

```

```

"encOpckKey": "hsm://mvno-beta/opc/imsi-416775586278902",
"encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278902",
"protectionParameterId": "pp-01",
"sequenceNumber": {
  "lastIndexes": {
    "ausf": 0
  },
  "sqn": "00000000020",
  "sqnScheme": "NON_TIME_BASED"
},
"supi": "imsi-416775586278902"
}
response:
{
  "algorithmId": "milena",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpckKey": "hsm://mvno-beta/opc/imsi-416775586278902",
  "encPermanentKey": "hsm://mvno-beta/k/imsi-416775586278902",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "00000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416775586278902"
}

```

PUT ndr /nudr-dr/v2/subscription-data/imsi-416775586278902/41677/provisioned-data/am-data -> 400

```

request:
{
  "gpsis": [
    "msisdn-141686278998"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
  "subscribedUeAmbr": {
    "downlink": "1 Gbps",
    "uplink": "200 Mbps"
  },
  "vendorSpecific-dxps": {
    "plan": "5G-100GB",
    "tenant": "mvno-beta"
  }
}
response:
{
  "cause": "MANDATORY_IE_INCORRECT",
  "detail": "simulated validation error",
  "status": 400,
  "title": "Bad Request"
}

```

Final order (GET /serviceOrder/{id})

```

{
  "@type": "ServiceOrder",
  "id": "44a48169-3aae-50a2-9e8a-455f61bc205d",
  "href": "/tmf-api/serviceOrdering/v5/serviceOrder/44a48169-3aae-50a2-9e8a-455f61bc205d",
  "priority": "1",
  "state": "failed",
  "orderDate": "2026-10-06T20:43:17.163208-07:00",
}

```

```

"completionDate": "2026-10-06T20:43:17.653163-07:00",
"serviceOrderItem": [
  {
    "id": "1",
    "action": "add",
    "service": {
      "serviceSpecification": {
        "id": "CreateSubscriber5G"
      },
      "serviceCharacteristic": [
        {
          "name": "plan",
          "value": "5G-100GB"
        },
        {
          "name": "supi",
          "value": "imsi-416775586278902"
        },
        {
          "name": "msisdn",
          "value": "141686278998"
        }
      ]
    },
    "state": "failed"
  }
],
"errorMessage": [
  {
    "@type": "ServiceOrderErrorMessage",
    "code": "DXPS-3400:MANDATORY_IE_INCORRECT",
    "reason": "item 1",
    "message": "rolled back after task failure"
  }
]
}

```

Appendix B - Network simulator payload catalogue

1,249 exchanges were harvested from the running simulators during a chaos load run (250 orders, 5/s, chaos 0.3). For each simulator: a successful exchange and, where one occurred, a fault-injected one. Credentials are redacted at the simulator.

Simulator	Exchanges	HTTP status	Injected faults
hub-sink	476	204 x476	-
ims-pg	68	200 x68	-
nef	19	201 x19	-
npdb	9	200 x9	-
ocs-bss	125	201 x104, 204 x4, 400 x2, 409 x11, 503 x4	invalid x2, flaky-recovered x24, flaky x4
restconf	36	200 x29, 204 x3, 409 x4	-
smdp	108	200 x92, 503 x16	flaky-recovered x21, unavailable x12, flaky x4
udr	408	201 x280, 204 x87, 400 x8, 404 x6, 503 x27	flaky x27, flaky-recovered x22, invalid x8

B.1 hub-sink

Success: POST /hooks/mvno-alpha -> 204 (HTTP/2.0, tenant mvno-alpha, 22.9 ms)

```
request:
{
  "event": {
    "serviceOrder": {
      "externalId": "DEMO-MVNO-ALPHA-8030015",
      "id": "5eface22-fd52-564c-a1c2-5964bc2386f3",
      "state": "pending"
    }
  },
  "eventId": "01a11475-7dd5-7b7b-b137-1b0ce780a669",
  "eventTime": "2026-10-07T03:43:26.6777524Z",
  "eventType": "ServiceOrderStateChangeEvent"
}
```

B.2 ims-pg

Success: PUT /ims-pg/v1/subscribers/imsi-416773008030006/impu -> 200 (HTTP/2.0, tenant mvno-beta, 4.9 ms)

```
request:
{
  "barring": false,
  "impi": "imsi-416773008030006@ims.mnc077.mcc416.3gppnetwork.org",
  "impu": [
    "sip:+962738030006@ims.mnc077.mcc416.3gppnetwork.org",
    "tel:+962738030006"
  ],
  "serviceProfile": "VoNR-DEFAULT"
}
response:
{
  "resource": "impu",
  "result": "OK",
  "supi": "imsi-416773008030006"
}
```

B.3 nef

Success: POST /quality-on-demand/v1/sessions -> 201 (HTTP/2.0, tenant host-mno, 9.9 ms)

```
request:
{
  "applicationServer": {
    "ipv4Address": "198.51.100.10"
  },
  "device": {
    "phoneNumber": "+962718030027"
  },
  "duration": 600,
  "qosProfile": "QOS_E",
  "sink": "https://localhost:9109/camara/notifications"
}
response:
{
```

```

"applicationServer": {
  "ipv4Address": "198.51.100.10"
},
"device": {
  "phoneNumber": "+962718030027"
},
"duration": 600,
"qosProfile": "QOS_E",
"qosStatus": "AVAILABLE",
"sessionId": "qod-1791344608638342700",
"sink": "https://localhost:9109/camara/notifications"
}

```

B.4 npdb

Success: PUT /npdb/v1/numbers/962728030137 -> 200 (HTTP/2.0, tenant mvno-alpha, 6.6 ms)

```

request:
{
  "donorNetwork": "OPERATOR-X",
  "msisdn": "962728030137",
  "recipientNetwork": "mvno-alpha",
  "routingNumber": "RN-MVNO-ALPHA"
}
response:
{
  "msisdn": "962728030137",
  "routingNumber": "RN-MVNO-ALPHA",
  "status": "PORTED_IN"
}

```

B.5 ocs-bss

Success: POST /tmf-api/accountManagement/v5/billingAccount -> 201 (HTTP/2.0, tenant mvno-alpha, 5.5 ms)

```

request:
{
  "@type": "BillingAccount",
  "accountType": "prepaid",
  "characteristic": [
    {
      "name": "msisdn",
      "value": "962728030008"
    },
    {
      "name": "plan",
      "value": "5G-ALPHA-50"
    }
  ],
  "name": "acc-962728030008",
  "relatedParty": [
    {
      "@REFERREDType": "Organization",
      "id": "mvno-alpha",
      "role": "brand"
    }
  ]
}
response:
{
  "@type": "BillingAccount",
  "accountType": "prepaid",
  "characteristic": [
    {
      "name": "msisdn",
      "value": "962728030008"
    },
    {
      "name": "plan",
      "value": "5G-ALPHA-50"
    }
  ],
  "href": "/tmf-api/acc/acc-962728030008",
  "id": "acc-962728030008",
  "name": "acc-962728030008",
  "relatedParty": [
    {
      "@REFERREDType": "Organization",

```

```

    "id": "mvno-alpha",
    "role": "brand"
  }
],
"state": "active"
}

```

Fault / rejection: POST /tmf-api/accountManagement/v5/billingAccount -> 503 (HTTP/2.0, tenant mvno-beta, 5.4 ms, fault flaky)

```

request:
{
  "@type": "BillingAccount",
  "accountType": "prepaid",
  "characteristic": [
    {
      "name": "msisdn",
      "value": "962738030997"
    },
    {
      "name": "plan",
      "value": "LTE-20GB"
    }
  ],
  "name": "acc-962738030997",
  "relatedParty": [
    {
      "@referredType": "Organization",
      "id": "mvno-beta",
      "role": "brand"
    }
  ]
}
response:
{
  "cause": "TEMPORARILY_UNAVAILABLE",
  "detail": "simulated transient failure",
  "status": 503,
  "title": "Service Unavailable"
}

```

B.6 restconf

Success: PATCH /restconf/data/ietf-l3vpn-svc:l3vpn-svc -> 200 (HTTP/2.0, tenant ent-acme, 30.7 ms)

```

request:
{
  "ietf-yang-patch:yang-patch": {
    "edit": [
      {
        "edit-id": "site-1",
        "operation": "merge",
        "target": "/ietf-l3vpn-svc:l3vpn-svc/sites/site=SITE-30016",
        "value": {
          "ietf-l3vpn-svc:site": [
            {
              "service": {
                "qos": {
                  "svc-input-bandwidth": 1800,
                  "svc-output-bandwidth": 1800
                }
              },
              "site-id": "SITE-30016",
              "site-vpn-flavors": {
                "site-vpn-flavor": "site-vpn-flavor-single"
              },
              "tenant": "ent-acme",
              "vpn-policies": {
                "vpn-policy": [
                  "@{entries=System.Object[]; vpn-policy-id=ACME-CORP}"
                ]
              }
            }
          ]
        }
      ]
    }
  },
  "patch-id": "01a11475-7d2f-74c4-845b-8f81e37ab683"
}

```

```
... (truncated)
response:
{
  "ietf-yang-patch:yang-patch-status": {
    "ok": [
      null
    ],
    "patch-id": "01a11475-7d2f-74c4-845b-8f81e37ab683"
  }
}
```

Fault / rejection: PATCH /restconf/data/ietf-l3vpn-svc:l3vpn-svc -> 409 (HTTP/2.0, tenant ent-acme, 15.5 ms)

```

request:
{
  "ietf-yang-patch:yang-patch": {
    "edit": [
      {
        "edit-id": "site-1",
        "operation": "merge",
        "target": "/ietf-l3vpn-svc:l3vpn-svc/sites/site=SITE-30030",
        "value": {
          "ietf-l3vpn-svc:site": [
            {
              "service": {
                "qos": {
                  "svc-input-bandwidth": 20000,
                  "svc-output-bandwidth": 20000
                }
              },
              "site-id": "SITE-30030",
              "site-vpn-flavors": {
                "site-vpn-flavor": "site-vpn-flavor-single"
              },
              "tenant": "ent-acme",
              "vpn-policies": {
                "vpn-policy": [
                  "@{entries=System.Object[]; vpn-policy-id=ACME-CORP}"
                ]
              }
            }
          ]
        }
      }
    ]
  },
  "patch-id": "01a11475-87a5-7809-9ff4-3c564a83122b"
... (truncated)
response:
{
  "ietf-yang-patch:yang-patch-status": {
    "edit-status": {
      "edit": [
        {
          "edit-id": "site-1",
          "errors": {
            "error": [
              {
                "error-message": "svc-input-bandwidth exceeds PE capacity (10000 Mbps)",
                "error-tag": "invalid-value",
                "error-type": "application"
              }
            ]
          }
        }
      ]
    }
  },
  "patch-id": "01a11475-87a5-7809-9ff4-3c564a83122b"
}
}

```

B.7 smdp

Success: POST /gsma/rsp2/es2plus/downloadOrder -> 200 (HTTP/2.0, tenant host-mno, 45.0 ms)

```
request:
{
  "eid": "89049032100000000000000008030004",
  "header": {
```

```

    "functionCallIdentifier": "01a11475-7c76-75f0-813d-1d879ac3f73b",
    "functionRequesterIdentifier": "host-mno"
  },
  "profileType": "5G-CONSUMER"
}
response:
{
  "header": {
    "functionExecutionStatus": {
      "status": "Executed-Success"
    }
  },
  "iccid": "8949000000008030004"
}

```

Fault / rejection: POST /gsma/rsp2/es2plus/downloadOrder -> 503 (HTTP/2.0, tenant mvno-beta, 30.0 ms, fault flaky)

```

request:
{
  "eid": "890490323000000000000000008030997",
  "header": {
    "functionCallIdentifier": "01a11475-8de3-74e3-93bb-e2e8ce0b537c",
    "functionRequesterIdentifier": "mvno-beta"
  },
  "profileType": "5G-CONSUMER"
}
response:
{
  "cause": "TEMPORARILY_UNAVAILABLE",
  "detail": "simulated transient failure",
  "status": 503,
  "title": "Service Unavailable"
}

```

B.8 udr

Success: PUT /nudr-dr/v2/subscription-data/imsi-416771008030003/authentication-data/authentication-subscription -> 201 (HTTP/2.0, tenant host-mno, 22.6 ms)

```

request:
{
  "algorithmId": "mileneage",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpkKey": "hsm://host-mno/opc/imsi-416771008030003",
  "encPermanentKey": "hsm://host-mno/k/imsi-416771008030003",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "000000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416771008030003"
}
response:
{
  "algorithmId": "mileneage",
  "authenticationManagementField": "8000",
  "authenticationMethod": "5G_AKA",
  "encOpkKey": "hsm://host-mno/opc/imsi-416771008030003",
  "encPermanentKey": "hsm://host-mno/k/imsi-416771008030003",
  "protectionParameterId": "pp-01",
  "sequenceNumber": {
    "lastIndexes": {
      "ausf": 0
    },
    "sqn": "000000000020",
    "sqnScheme": "NON_TIME_BASED"
  },
  "supi": "imsi-416771008030003"
}

```

Fault / rejection: PUT /nudr-dr/v2/subscription-data/imsi-416773008030002/41677/provisioned-data/am-data -> 400 (HTTP/2.0, tenant mvno-beta, 3.1 ms, fault invalid)

```

request:
{
  "gpsis": [
    "msisdn-962738030998"
  ],
  "internalGroupIds": [
    "mvno-beta"
  ],
  "nssai": {
    "defaultSingleNssais": [
      {
        "sd": "000001",
        "sst": 1
      }
    ]
  },
  "ratRestrictions": [],
  "subscribedDnnList": [
    "internet"
  ],
  "subscribedUeAmbr": {
    "downlink": "1 Gbps",
    "uplink": "200 Mbps"
  },
  "vendorSpecific-dxps": {
    "plan": "5G-100GB",
    "tenant": "mvno-beta"
  }
}
response:
{
  "cause": "MANDATORY_IE_INCORRECT",
  "detail": "simulated validation error",
  "status": 400,
  "title": "Bad Request"
}

```

Appendix C - Deviations from the design in this build

Design	This build	Reason
Protobuf on Kafka	JSON with versioned contract structs	Readable payload evidence; schema registry out of scope locally
NE selection via NRF discovery	Registry lookup by ne:{nfType}; NRF used for OAuth tokens	Single simulator per NF type
Kafka SASL_SSL	PLAINTEXT bound to 127.0.0.1	Single-node dev runtime; PostgreSQL, gateway and NE links use TLS/mTLS
Kafka compaction and retention	Disabled in the Windows dev runtime	Kafka on Windows cannot rename open segments (KAFKA-1194)
Cross-domain atomicity	Atomic per business command (saga per BC)	As designed for the MVP; cross-BC atomicity via order-level policy
eSIM asynchronous callbacks (ES2+ handleDownloadProgressInfo)	Synchronous confirm/cancel	Simulator scope
go test -race	Not run	Requires cgo, which is unavailable on this machine
NE latency SLOs	Not asserted	All components share one laptop; see section 8